(1)

DMMM - Disaster management & Mitigation measures

Natural   Man-made.

Definition of Disaster :

(2) A Disaster can be defined as an occurance either natural or manmade that has an environmetal and socio-Economic ~~effect~~ effect on people in terms of loss of life, loss of property and any other significant loss.

(3) Disaster is a serious disruption of the functionality of a community or a society which involves loss of human life, material, economic and environmental losses and impact which exceeds the ability of the affected community or society to cope; using its own resources.

(1) A disaster is a concequence of naturally occuring phenomenon. or manmade event that cause damage, Ecological - disruption, loss of human life, deterioration of health & health services.

Disaster management : It comprises of both Risk-(reduction) management & crisis management.

Risk management includes : ① prevention & mitigation
② preparedness
③ predicting and early warning.

crisis management : ① Response
② Relief & Rehabilitation
③ Impact assesment.

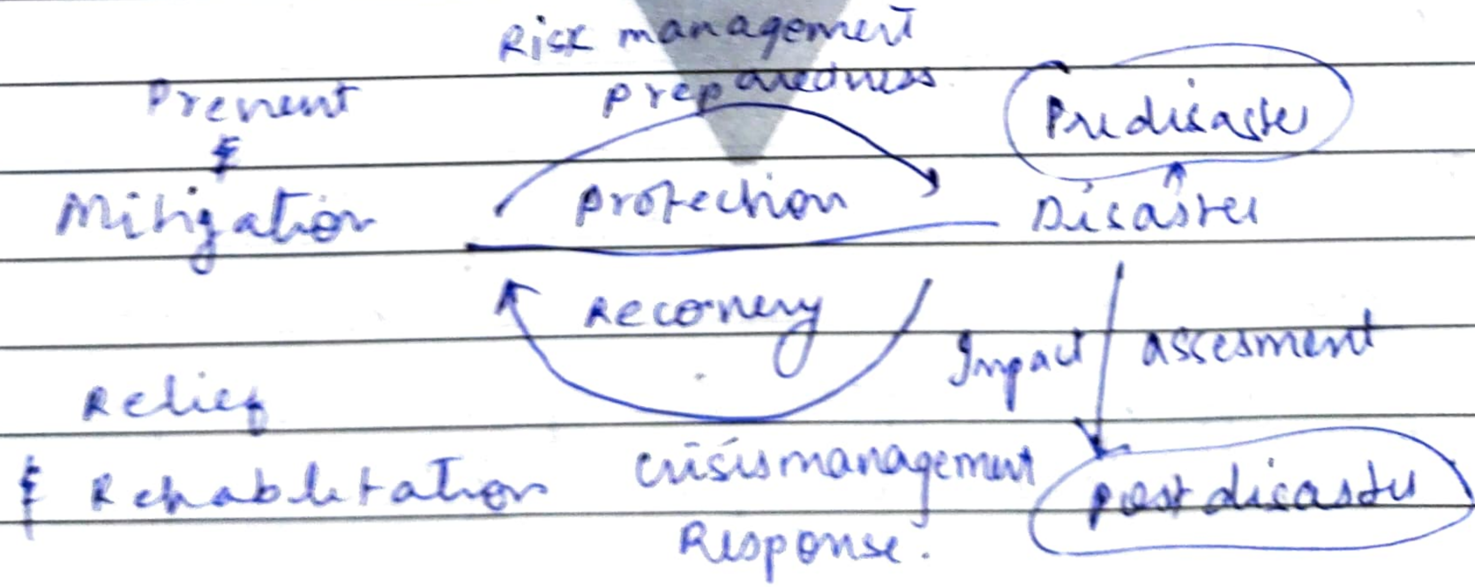


fig 1 : flow chart of disaster management

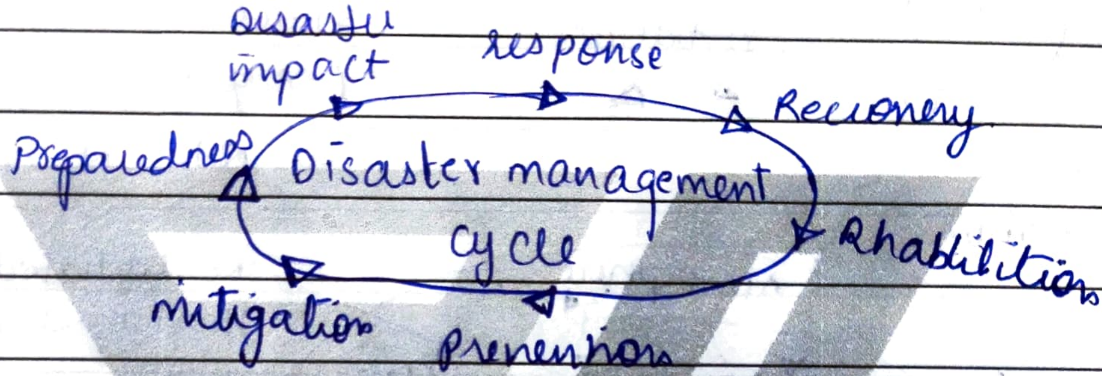


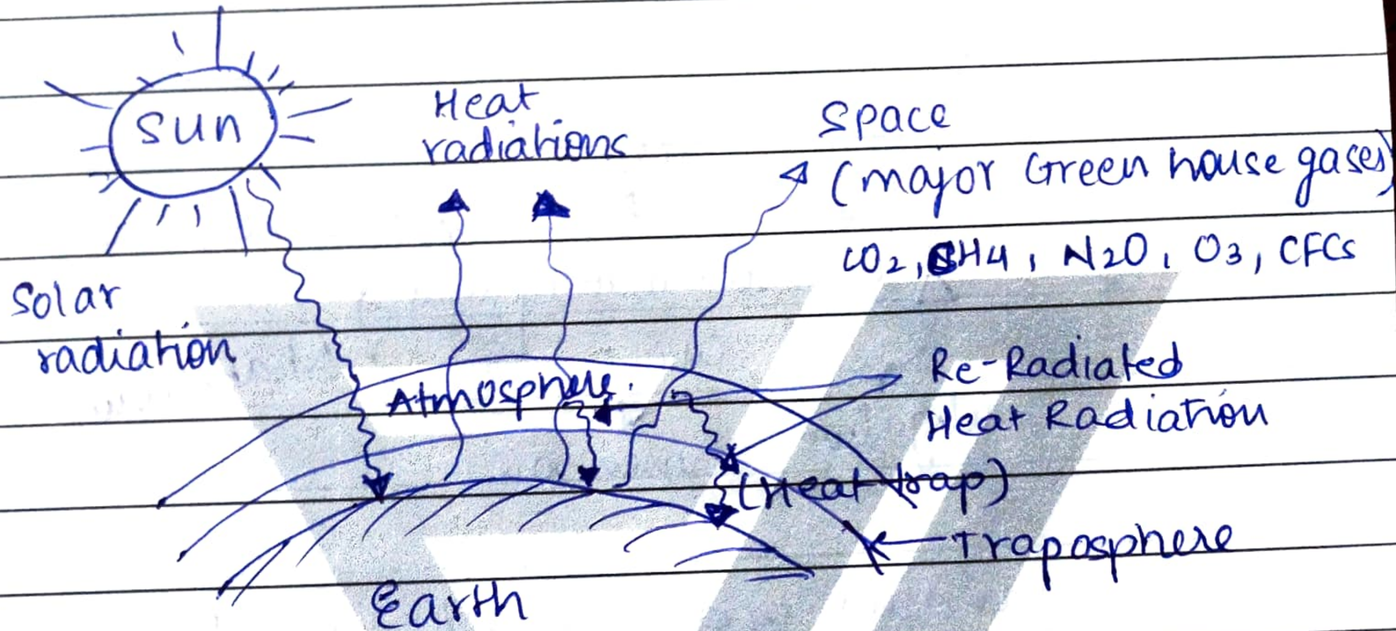
fig: a standard cycle of Disaster Events.

Defenition of HAZARD

- Hazard is a probablity of occurence of a potentially damaging phenomenon or threatning-event within a given time period and area.
- Hazard can only becomes disaster when it adversly affects the human society.

Example
- Biological Hazards: cholera, dengue-fever, Malaria, Tuberculosis etc.
- Metereological: cyclone, storms, Tidal waves, Hurricane etc.
- Technological Hazards: Transport Accident.

④ Green house effect & global warming.



- Troposphere is the lowest layer of the atmosphere which traps heat by a natural process, due to the presence of certain gases [green-house gases] such as CO_2, NH_4, N_2O_3, CFCs etc] Heat trapped by green house gases keeps the planet (earth) warm enough to allow us and other species to exist. This effect is called Green house effect.

- Deforestation [cutting of plants] has resulted in elevated (increased) levels of CO_2 due to non-removal of CO_2 by plants through photosynthesis. Other gases whose levels have increased due to human activities, methane (CH_4), nitrous oxide (N_2O) & chlorofluoro carbon (CFCs).

- Due to anthropogenic activities (man-made/man induced) includes increase in the concentration GHG's in the air that absorbs Infrared (IR) light containing heat. This results in reradiation of even more outgoing Thermal IR-energy; there by increasing the average surface temperature of earth beyond 15°C (the average). This phenomenon is reffered as enhanced green house effect or global-warming.

The more and more increase in the tepr of air & the earth. This increase in the average global temp on the earth is called global warming.

DMMM

- Introduction to Global warming & climate change.
- climate is the average weather of an area.
- climate is the general weather conditions, seasonal variations and extreme of weather in a region (area). Such weather conditions which average over a long period (at least 30) years is called climate.
- Extreme change in weather (climate change) is mainly to anthropogenic (man-made) activities which are responsible for upsetting the delicate balance that has established between various components of the environment
- Extreme climate change is also due to Green house gases which are increase in the atmosphere resulting in increase in Global tempratures this may upset hydrological cycle. which results in floods and droughts in different regions of the world, also cause sea level rise changes in agricultural productivity, famines and death of humans as well as livestock
- It also causes disturbed rainfall, which results in some area wetter & the other drier.

Impacts and effects ← direct / Indirect.

Disaster occurs when hazard meets vulnerable situation

Stage 1 pollution → Stage 2 Hazard → Stage 3 vulnerable situation

pollution → Disaster ← vulnerable situation

- Example - Earth quake in Gujrat in 2001
  flash flood in Uttarakhand
  covid-19 ← Biological hazard.

DATE

Disaster

1. Huge economic loss
2. Disturbance or Disruption of people.
3. food and water Scarcity.
4. can alter the natural environment
5. psychological Impact
6. Increase the level of vulnerability

- Direct effects of disaster refers to direct quantifiable losses that is physical and structural impact caused by the disaster such as damage to property and destruction of property of cultural significance.

Direct effect of disaster

1. change in climate
2. change in Economy
3. loss of life
4. loss of Environmental Resources
5. Damage & Destruction of property.
6. Impact on human health

- Indirect effects of disaster refers to indirect non-quantifiers losses that is non-physical impact caused by the disaster such as disruption of essential goods and services.

Indirect effects of disaster

1. Loss of vegetation & Habitat
2. Loss of biodiversity
3. Loss of tourism
4. Loss of Jobs

- Change in weather
- Disruption of essential Goods and Services

Ozone Layer depletion

1) $O_2 + h\nu \text{ (photon)} \rightarrow 2O$

An oxygen molecule (O_2) reacts in the presence of photon ($h\nu$) from solar radiations & forms two oxygen ($2O$) atoms [atomic oxygen (O)]

2) $O + O_2 \rightarrow O_3$.

The atomic oxygen (O) rapidly reacts with molecular oxygen (O_2) to form ozone (O_3 - gas) Ozone (O_3) thus formed absorbed harmful ultraviolet radiation and is beigng continously converted back to molecular oxygen

$$O_3 + h\nu \longrightarrow O_2 + O.$$

Short term effects and long term effects of

- Short term effects are the immediate consequnses experienced during the disaster and immediatly after a disaster. (typically within first few days or weeks).

- short term effects includes loss of life, injuries, and significant damage to buildings and infrastructure.

- Some of the short term effects includes

Physical impacts	Psychological	Economic	Environmental
- loss of life & injuries - Damage to infrastru (such as bldgs, rods bridges etc). - Disruptions to essential services (clean water, food, electricity) Health care & communication	- PTSD. - shock, fear & confusion - feelings of helplessness and uncertainity. - Accute stress disorders (Anxiety, depression & sleep disturbances) - potential impacts long term mental health issues.	- disruptions to livelyhoods - Business can be destroyed, jobs lost etc - inc costs for recovery and re-building - food shortages	- pollution and contamination erosion and landslides.

Long term effects of disasters.

- emerged after the initial weeks an can persist for several years.
- can include
① mental health problems. like anxiety and depression
② as well as physical health problems, economic disruption and social capital decline.

1. mental health impacts
   - PTSD, depression or anxiety

2. physical Health
   - increase risk of suicide, emotional distress and avoidance behaviour
   - increase risk of injection, resperatory and gastro.
   - food scarcity and malnutrition, increase risk of cardio-vascular diseases

3. Economic & social
   - Economic disruption or damage
   - Job loss and displacement.
   - financial instability
   - Strain of infrastructure and services.
   - Increased community violence

4. Vulnerability and existance
   - preexisting ~~and~~ mental & physical health conditions become more and more vulnerable

PTSD (Post Traumatic Stress disorder)

Addressing Long term effects

1) Providing mental health services

2) Economic Recovery programs which includes Creating jobs, providing financial Assistance which can help communities to recover economically

3) Investing in infrasture including health care facility, Sanitation systems, & transportation networks.

DATE

Module 2: DMMM

natural Disaster and manmade disaster

Diffrence between Hazard and disaster.

Hazard	Disaster
- A Hazard is a perceived natural-event which threatens life, property and environment	- A disaster is consequence of hazard which occurs when hazard meets vulnerable situation leads to great loss of life, prop and e-commerce
- community and e-commerce is able to cope with hazard using available resource	- A community is overwhelmed by hazard and is unable to manage with current resources
- Initial - hazard is not considered as disaster	- Hazard leads to disaster by creating vulnerable situation

Hazards

Disasters

Technological hazards.

$\boxed{\text{Natural Hazards}}$

- Hydrometeorological disaster
  - floods, flashfloods, cyclones, cloud burst, storm, storm-surge, Extreme temparatures
- Geological - disaster
  Earthquake, tsunamis, volcanic erruptions, Landslides

Environmental degradation

Biological disasters
- Epidemic such as covid-19.

Pollution (Stage 1) → Hazard (Stage 2) → Vulnesituation (State 3)

Disaster

Q) Explain the following disaster with respect to preparedness (1), response (2), recovery (3) and mitigation (4) (10 mrks)

(1) floods (2) Earthquake (3) Landslides (4) cyclones (5) Tsunamis.

— (2) Earthquake:

- An earthquake is a sudden shaking of the ground caused by the movement of tectonic plates beneath the earths surface.
- Earthquake can cause severe dmg to buildings, infrastruc and can result in loss of life, injuries and disruptions of daily life.
- Earthquake can be understood more effectively when analyzed through the four phases of disaster managentment i.e preparedness, response, recovery, mitigation
- These four phases form a cycle that helps the communities to reduce the risk and to manage the crisis effectively and to built resilience

① preparedness : Actions taken in advance before an earth quake to minimize the damage and protect lifes

- Key actions :
① Education and awareness
② public awareness campaings about what to do before, during and after an earthquake
③ Training and drills.
④ Teaching the communities how to drop, cover and hold during earthquake.
⑤ emergency kits : emergency planning, encouraging families to prepare supplies like water, food, first ait and flaslight.
⑥ Building codes : enforcing a seismic plate to ensure or make building more earthquake resistance to ensure Safety and minimize loss of life and property and provide emergency aid

② Response !

Key actions :
① Search an rescue : locating and saving trapped or injured people.
② Setting emergency medical camps and provide treatment to the injured
③ shelter and relief distribution, providing temporary shelters. distributing food, water and aid.
④ communication and coordination : provide service coordination and emergency services and maintaining communication between authorities ~~&~~ rescue teams and

public

③ Recovery: The process of restoring the normal life over the weeks or months after the earthquake

Key actions:

① Reconstruction of infrastructure: eg Repairing and rebuilding schools, roads etc

② economic Supporting and helping families to resume their business and to recover financially.

③ Psychological Support: Mental health support, providing counseling to the Survivors.

④ restoring services: Bringing back utilities like water and transportation

④ Mitigation: Reducing the long term risk and impact of future earthquake

- Retrofitting: Upgrading old buildings or strengthening to meet Safety Standards. and to resist Seismic forces.

- Land used planning: avoiding construction on high risk Sesamic zones or unstable grounds.

- Early warning System: Investing and improving the alert System.

Q] What is between Storm and Storm surge.

- A cyclone is a weather system characterized by strong winds heavy rainfall and flooding and is often accompanied by a storm surge.

- Storm Surge (flooding) (Storm tide) is a abnormal rise in sealevel which is caused by a cyclone and it leads to pushing water in land (costal flooding.

- Essentially, the cyclone is the storm itself. While, a storm surge is a dangerous costal ~~st~~ flooding effect caused by the cyclone.
- Depending on the region, cyclones are also called, typhoons. in short a storm surge is an abnormal rise in sea level caused by a cyclone.

- causes of storm surge.
  - It is primaraly caused by strong winds pushing the water towards the coast. and at the low atmospheric pressure associated with the cyclone.

- Impact of Storm Surge
  - Can cause wide spread coastal flooding, causing significant Dmg to property, infrastucture and loss of life.

- The intensity of the storm surge depends upon strength of the cyclone, specially the ~~wid~~ wind speed near the storm center.

- The strongest winds in a cyclone are typically found in eyewall and storm surges tends to be more intense in that area.

- Eyewall : it is the area with the strongest winds of storm and heaviest rainfall are typically formed.

Q] Explain "cloud-burst".

- Acc to IMD cloud burst involves rainfall of 10 cm (100 mm) or more within an hour over a small area of 10 square km to 30 ~~Squa~~ Sq/km.

- Sudden, localized and intense rainfall caracterized by a heavy downpour of rain water over a relatively small area. in a short period of time

~~Information~~

DMMM Disaster Management [30th July

- man-made ~~register~~ disaster

1. chemical Hazard — role of growing population
Industrial — Industrialization
Nuclear — urbanisation
fire — changing lifestyle of human beign

} in frequent occurance of manmade

2 marks

/ Manmade disaster: It is an event that causes significant damage, destruction or loss of life due to human actions or human negligence.

- man made disaster are caused by human actions weather intentional or accidental.
- Examples. ① Bhopal gas tragedy.
  - A massive leakage of toxic gas, methayl iso cynide from pesticides in 1984

② Chernobyl disaster:
  - A nuclear accident in Ukraine in 1986, that release radio active materials into the environment

③ 9/11 terrorist attacks on world trade center.
  - And pentagon. resulting in wide spread destruction and loss of life.

10 marks

Q) Explain the following disaster w.r.t. prepared., response
recovery and mitigation (iv) fire — "

Before disaster — prepared; During disaster — response; After disaster — recovery; mitigation (Reducing future risk)

(i) Industrial Hazard.
(ii) Nuclear — "
(iii) chemical — "

DATE

- Industrial disaster / Hazard
- Refers to a major accident originating from industrial activities involving hazardous materials, chemicals machinary failure, explosion gas leaks or fires.
- These disaster can lead to large scale damage to human health, environment, loss of life and property.
- Example: Bhopal gas Tragedy, 1984 in India

— 1) preparedness:
- Objective: is to reduce vulnerability and enhance the capacity to respond a readiness.
- Key actions / activities
  - i) Emergency plans
    - develop and communicate on site and offsite emergency response plans and evacuation routes.
  - ii) Training and drills
    - conduct safety drills a mock drills and train the workers in emergency response procedure
  - iii) Safety equipments:
    - Install gas detectors, fire extinguishers, alarms and emergency exits c
  - iv) coordination
    - establish communication with local emergency services (hospitals, fire brigades etc).

— 2) Response (during the disaster)
- objectives
  To manage the emergency effectively to save the lives and minimize damage.

- Key action | activities
i) activate emergency protocols
- sound of alarms
ii) Rescue operations
- Mobilise fire services, medical teams, Hazmat squats.
iii) medical care
- provide first aid and evacuate injured to health care facilities.
iv) containment:
- Shut down machinary, isolate chemicals and stop the spread of Hazard

— 3) Recovery:
Objective:
- To restore normaly and rehabitalet affected areas and people
- Key actions
i) Medical rehablitation
- offer continued treatment to affected individuals.
2) Environmental clean up.
- de contominate soil, water & air
3) damage assesment.
- Evaluate Economic and physical loss.
4) reconstruction
- Repair and rebuild infrastructure and resume operations
5) legal and finatcial support
- compensation to victims and legal accountability

— 4) Mitigation
- Objective:
Is to prevent or reduce the impact of future disasters.

- Key actions / activities.
- ① strict regulation.
  - Enforce industrial safety laws and inspections
- ② Safer technologies
  - upgrade equipment, install fail safe system and automatic controls.
- ③ Buffer zones.
  - Avoid placing industries near residential areas.
- ④ community awareness.
  - Educate nearby populations about risk and evacuation plans.
- ⑤ Risk insurance
  - secure insurance coverage for disaster related losses

☆ ⇒ Q] Role of growing population as manmade disaster in India

- Uncontrolled & rapid population growth in india is considered to be a manmade disaster due to its negative impacts on society, economy and infrastructure.
- population explosion creates pressure that leads to various secondary disaster which are human induced.
- In case of population explosion the causes are, lack of family planning, low eduction, and cultural factors which are human influenced.
- The effects of growing population are,
  - unemployment
  - resource depletion etc.

which leads to large scale distress

- Impacts of population growth in India.

1. Environmental degradation.
   - i) air and water pollution
     Over population in urban areas. causes untreated sewage, vehicular and industrial emmisions.
   - ii) climate impact.
     Increased carbon footprint due to industrial and human activities
   - iii) Deforestation.
     forests are cleared for housing and agriculture.

2. Strain on resources:
   - i) Water scarcity:
     Overuse of ground water river pollution and unequal access/ distribution.
   - ii) Food insecurity
     Agricultural land shrinking under urban pressure
   - iii) Energy crisis.
     High demand for electricity fuel and energy sources.

3. Urbanisation pressure.
   - i) Slums and overcrowding.
   - ii) inadequate infrastructure
   - iii) Health crisis
     Spread of diseases, lack of access to health care.
   - iv) unemployment
     Job shortage, child labour and migration
   - v) education ~~an~~ deficit.

DMMM [31st July]

#

→ Disasters are classified into 2 types.

Natural Disaster

Geophysical	Hydrological	Meteorological & Climatological
• Earthquake	• floods	• storm
• Tsunami	• flash-flood	• storm surge
• volcano	• mass-movement wet	• cyclone
• mass-mov-dry	• cloud burst	• Drought
• Land-slides	• avalanches	• wild-fire
	• Mud flow	

Manmade Disaster

Environmental	Technological
• Global warming	• chemical hazard
• Deforestation	• Nuclear
• climate change	• Industrial
• Ozone depletion	• fire hazards.

May come

Q) Discuss the changing lifestyle of human beings as ~~an~~ man-made disaster in India?

⟹ The rapidly changing lifestyle of human beings in india marked by urbanization, consumerism, dietary changes, sedentary habits. and technological dependencies. has led to several adverse effects. that resemble the Scale and impact of manmade disaster.

• These lifestyle changes often considered modern, contribute to health crisis ~~and~~ environmental degradation and socio economic imbalance.

- A manmade disaster is caused by human actions or negligence, the changing lifestyle in india driven by urban consumer culture, fast technology & Globalisation

border
- ~~cross water~~ trade in and it leads to rising health issues, waste generation and environmental harm. Which create long term irreversible changes in society and ______

Mitigation Measures.
- promote sustainable living
- Health awareness campaing.
- Green Urban planning
- Re-inforce Traditional Wisdom
- Digital detox & Tech balance.